Practice workbook · Explain validity problems and fixes in investigations on people

Investigations using volunteers or patients where the student explains why results are not valid, why a rest or fasting step improves validity, or what to change.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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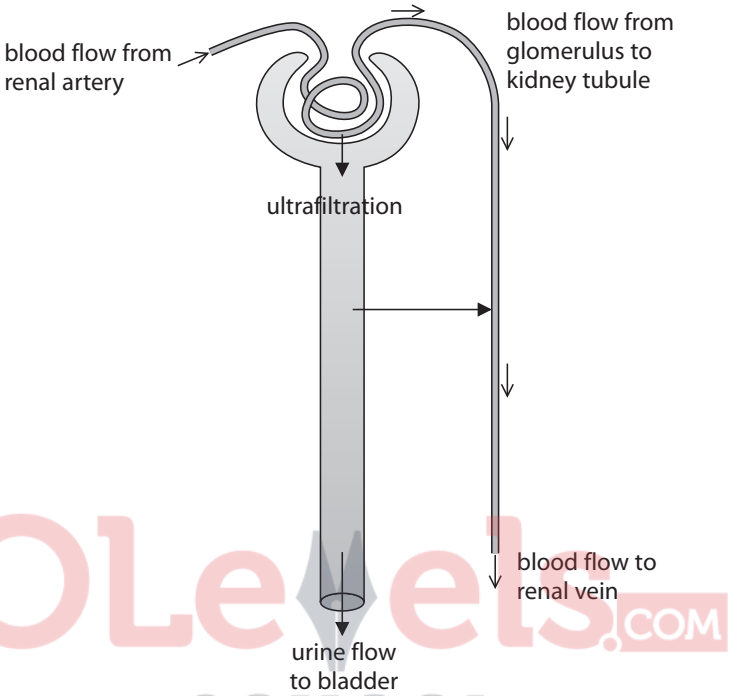
Type ID: HBI-PRAC-CONTROL-HUMAN-SUBJECT-VALIDITY

1. Question 8 · part b.ii · 2 marks

June 2019 Paper 2

Worked-example retry · Attempt part b.ii only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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1. Question 8 · part b.ii · 2 marks

June 2019 Paper 2

Worked-example retry · Attempt part b.ii only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

(4)

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

(1)

(ii) Explain why this investigation is unlikely to produce valid results.

(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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1. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

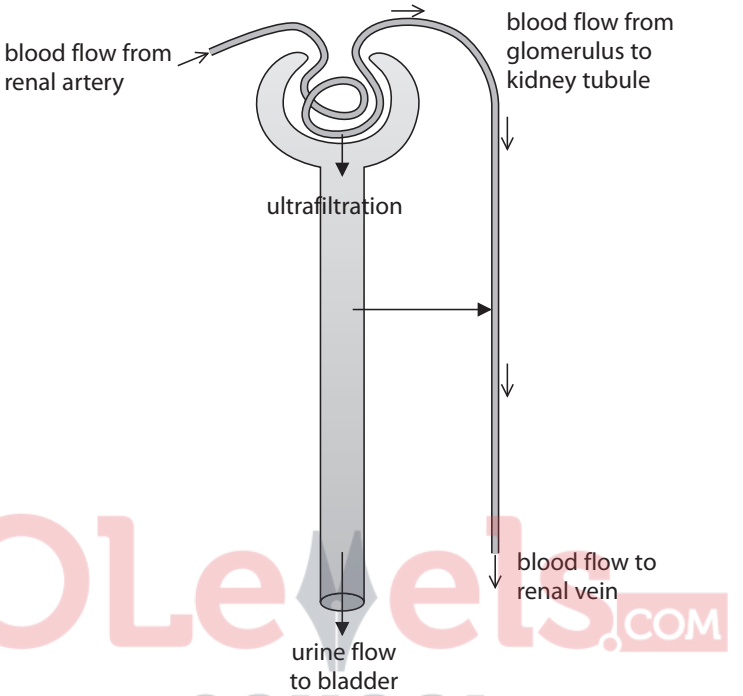
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2. Question 8 · part b.iii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.iii only.

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(Source: adapted from © Alila Medical Images/Alamy)

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2. Question 8 · part b.iii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.iii only.

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2. Question 8 · part b.iii · 1 mark · Working and answer

Working and answer

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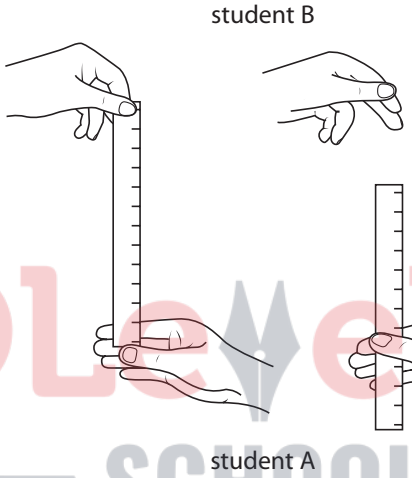
3. Question 5 · part d · 2 marks

November 2021 Paper 2

Further practice · Attempt part d only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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3. Question 5 · part d · 2 marks

November 2021 Paper 2

Further practice · Attempt part d only.

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The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

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(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

3. Question 5 · part d · 2 marks

November 2021 Paper 2

Further practice · Attempt part d only.

(d) The student who catches the ruler does not eat or drink anything for several hours before the investigation.

Suggest why this improves the validity of the investigation.

(2)

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(e) A cup of coffee contains 100 mg of caffeine. A can of cola contains 50 mg of caffeine.

If the student catching the ruler had drunk a can of cola instead of coffee, estimate the mean distance travelled by the ruler.

(1)

mean distance = ... mm

(Total for Question 5 = 9 marks)

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Working and answer

4. Question 5 · part c.i · 2 marks

June 2022 Paper 2

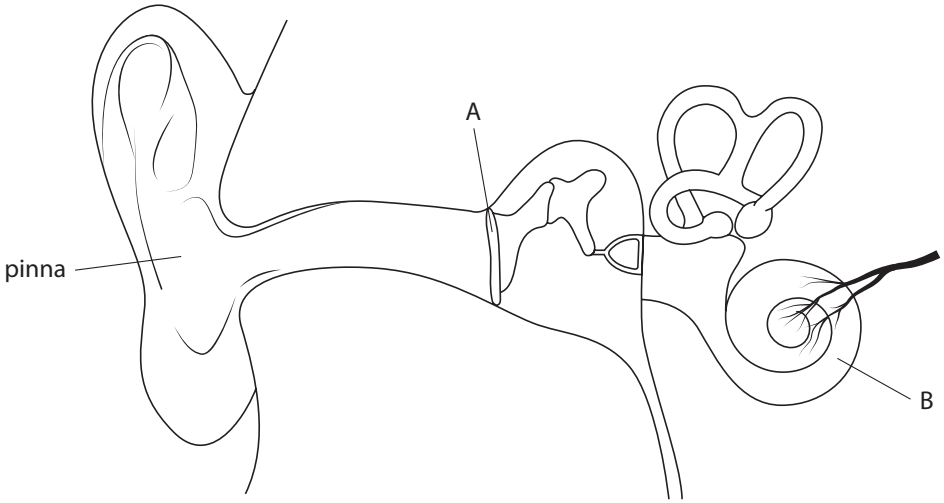
Further practice · Attempt part c.i only.

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5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

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4. Question 5 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

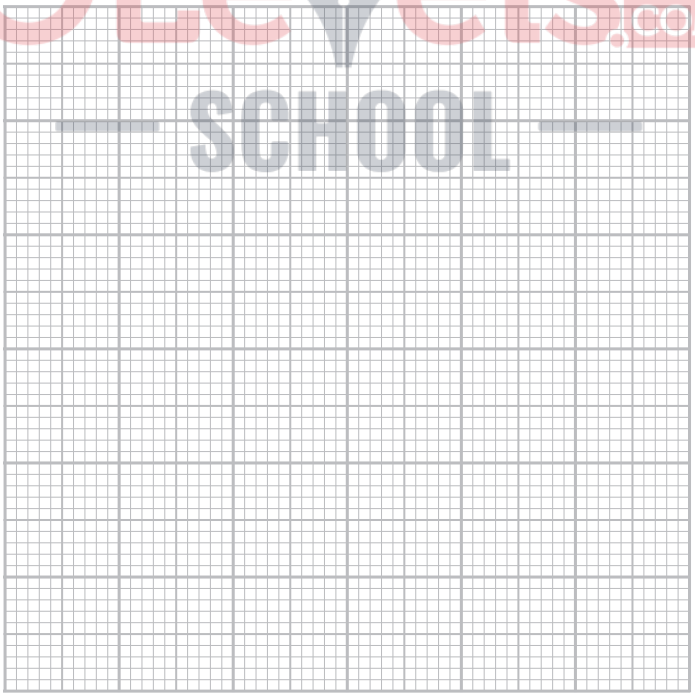
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)



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P 7 0 9 7 4 A 0 1 0 2 0

4. Question 5 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

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(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

(3)

percentage change = ... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

(2)

1 .....

.....

2 .....

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(ii) Describe how the reliability of the results could be improved.

(2)

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(Total for Question 5 = 15 marks)

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4. Question 5 · part c.i · 2 marks · Working and answer

Working and answer

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5. Question 6 · part b.iv · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iv only.

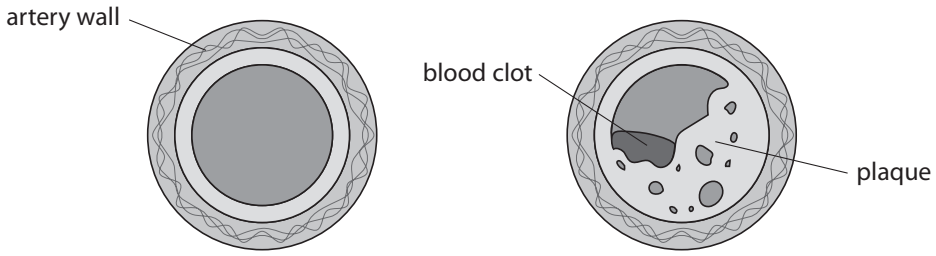
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6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery

artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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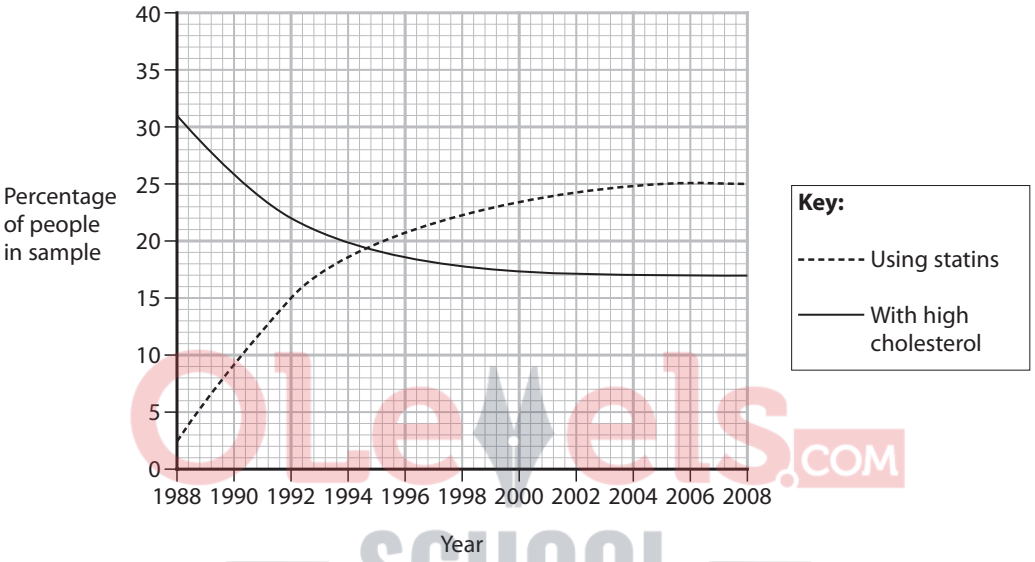
5. Question 6 · part b.iv · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iv only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

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(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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5. Question 6 · part b.iv · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iv only.

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(iii) Describe a method that scientists could use to obtain the data shown in the graph.

(5)

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(iv) Give two factors the scientists need to consider to make sure that their investigation is valid.

(2)

1 .....

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2 .....

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(Total for Question 6 = 15 marks)

5. Question 6 · part b.iv · 2 marks · Working and answer

Working and answer

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6. Question 5 · part a.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part a.i only.

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5 A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 cm^2 area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

(2)

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(ii) State three variables that should be controlled.

(3)

1 .....

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2 .....

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3 .....

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6. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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